Layer-Adaptive Sparsity for the Magnitude-based Pruning

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arXiv:2010.07611

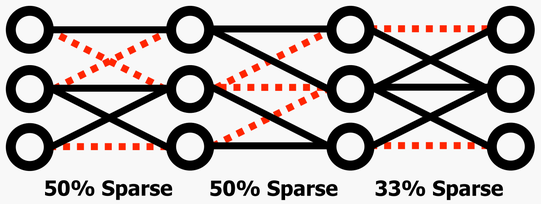
github.com/jaeho-lee/layer-adaptive-sparsity

TL;DR. We devise a layerwise sparsity ratio for magnitude-based pruning algorithms, based on a distortion-centric perspective (and it works well).

Preliminaries. MP and layerwise sparsity

An essential but not-so-highlighted component of pruning algorithms.

Pruning? An act of removing connections from a (big) neural network, usually for reducing storage/computation load.



Magnitude Pruning? Magnitude(-based) pruning (MP) is a simple method: Remove the connections with smallest magnitude in each layer!

Layerwise Sparsity? How sparse should each layer be? (Obviously, we would not want any layer to be fully-sparse!)

Motivation. Magnitude rules, but…

Only with a very-well-tuned layerwise sparsity.

LeCun et al. (1989) Proposed one of the first pruning algorithms (called OBD), and showed that OBD beats MP.

Han et al. (2015) Showed that MP performs sufficiently well for deep networks; MP can remove ~90% of weights without degrading performance.

Gale et al. (2019) Reported that MP is actually SOTA with good hyperparam. tuning, especially the right layerwise sparsity!

Do we really need to hand-tune the layerwise sparsity? (The complexity grows exponentially with depth!)

Baselines. What are being used?

Many rules, but no "standard" method.

Uniform. Every layer is pruned to the same sparsity level. (70% from the first layer, 70% from the second layer, …)

Uniform+. Same as Uniform, but uses following constraints:
(1) Keep the first convolutional layer unpruned.
(2) In the last layer, keep at least 20% of weights unpruned.

Global. Apply a "global" threshold (instead of a layerwise one), so that the layerwise sparsity is automatically tuned.

Erdös-Rényi Kernel (ERK). Makes each pruned layer look like an Erdös-Rényi random graph generated with same "connection probability" parameter.

All these are used, but there is no "go-to" method!

Approach. Layerwise MP = distortion min.

Viewing MP as a distortion-minimization procedure gives rise to LAMP.

Framework. We view MP as an ℓ_2 distortion minimization procedure: Consider a neural network taking a form

$$f(x; W_{1:d}) = W_d \sigma(W_{d-1}\sigma(\cdots W_2 \sigma(W_1 x)\cdots))$$

where $W_1, \ldots, W_d$ are weight matrices and $\sigma(\cdot)$ is ReLU.

Layerwise. Layerwise MP can be viewed as (approximately) minimizing the "distortion from pruning only one layer, w.r.t. worst-case input." That is, minimizing

$$\sup_{\|x\|\le 1} \|f(x; W_{1:d}) - f(x; W_{1:i-1}, \widetilde{W}_i, W_{i+1:d})\|_2^2$$

Indeed, one can "peel off" $W_{i+1:d}$ and consider

$$\sup_{\|\zeta\|\le 1} \|W_i\zeta - \widetilde{W}_i\zeta\|_2^2 = \|W_i - \widetilde{W}_i\|_2^2$$

where ζ is the input to the i-th layer.

MP minimizes the Frobenius distortion $\|W_i - \widetilde{W}_i\|_F^2$, which is almost equivalent to the spectral distortion.

What layerwise sparsity minimizes the "model distortion"?

$$\sup_{\|x\|\le 1} \|f(x; W_{1:d}) - f(x; \widetilde{W}_{1:d})\|_2^2 \quad (*)$$

Method. LAMP = model distortion min.

Viewing MP as a distortion-minimization procedure gives rise to LAMP.

LAMP. LAMP applies a global threshold on a scaled version of magnitude: For each weight matrix W, assume an "ascending" index map satisfying $|W[u]| \le |W[v]|$ whenever $u < v$. Then, **LAMP score** is defined as

$$\text{score}(u; W) = \frac{(W[u])^2}{\sum_{v \ge u} (W[v])^2}$$

which is designed to minimize **(*)**.

Intuition. LAMP score measures the sq. magnitude of the target connection, normalized by the sq. magnitudes of all connections that survives at the moment when the target connection is going to be pruned. (i.e., normalizing after pruning each individual connection).

Property. Pruning according to the LAMP score is
(1) equivalent to performing MP with auto-tuned layerwise sparsity.
(2) invariant to weight-rescaling, e.g.,

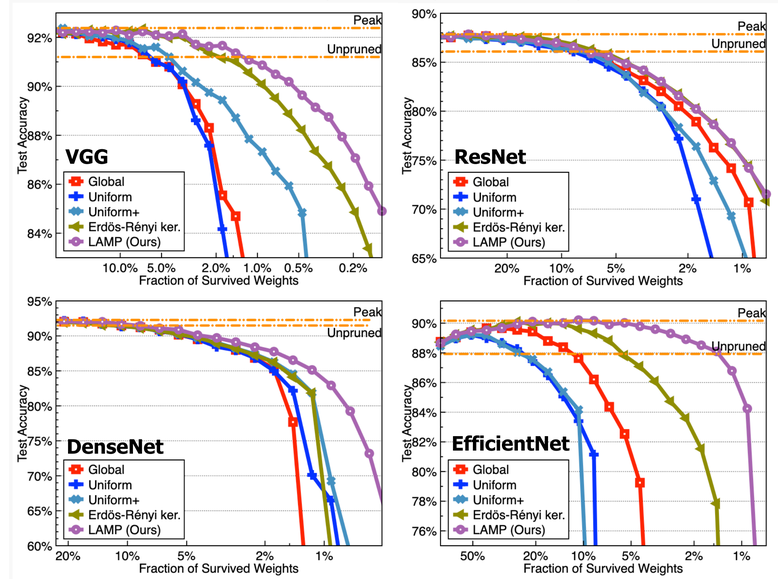
$$W_2\sigma\left(W_1 x\right) \mapsto \tau W_2 \sigma\left(\frac{1}{\tau} W_1 x\right)$$

Result. LAMP rocks.

LAMP is the best, ERK and Global is also good, uniform no.

Setup. We perform iterative MP, using AdamW with a fixed learning rate. Default model is VGG-16 (with batch norm), and dataset is CIFAR-10.

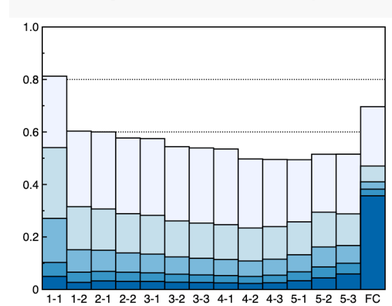
Models. We tested over VGG-16, ResNet-20, DenseNet-121, EfficientNet-B0



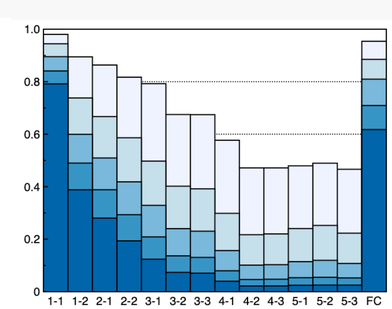
Datasets. LAMP also works well on: SVHN, CIFAR-100, Restricted ImageNet. (See Figure 3 of the paper!)

Others. We also test with different pruning schedules / rewinding. (See Section 4.2 of the paper!)

Heuristic Recovery. We observe that LAMP recovers some well-known handcrafted heuristics for the layerwise sparsity (cf. Uniform+).



Survival Rate of VGG-16 + Global



Survival Rate of VGG-16 + LAMP